FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Advanced), 2015

PART TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 210**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

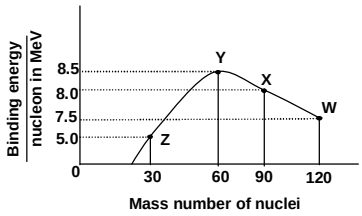
PART – I

SECTION – A

Straight Objective Type

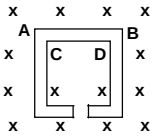
This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Binding energy per nucleon v_s mass number curve for nuclei is shown in the figure W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is



(A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$
(C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

2. A wire is bent to form the double loop shown in the figure. There is a uniform magnetic field directed into the plane of the loop. If the magnitude of this field is decreasing, current will flow from



(A) A to B and C to D (B) B to A and D to C
(C) A to B and D to C (D) B to A and C to D

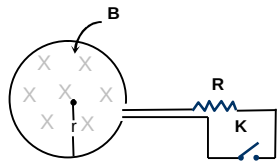
3. A cubical room is formed with 6 plane mirrors. An insect moves along the diagonal of the floor with uniform speed. The speed of its image in two adjacent walls is $20\sqrt{2}$ cm/s. Then the speed of the image formed on the roof is
(A) 20 cm/s (B) 40 cm/s
(C) $20\sqrt{2}$ cm/s (D) $10\sqrt{2}$ cm/s

4. A prism of refractive index $\sqrt{2}$ refracting angle A produces minimum deviation D_m of a ray on one face at an angle of incidence 45° . The values of A and D_m are respectively.
(A) 45°, 45° (B) 45°, 60°
(C) 60°, 30° (D) 60°, 45°

5. Which of the following statement is correct?
(A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.
(B) The rest mass of a stable nucleus is greater than the sum of the rest masses of separated nucleons.
(C) In nuclear fission, energy is released by fusing two nuclei of medium mass (approximately 100 amu)
(D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus.

Space for rough work

6. Shown in the figure is a circular loop of radius r and resistance R. A variable magnetic field of induction $B = e^{-t}$ is established inside the coil. If the key (K) is closed at t = 0, the electrical power developed in the external resistor at this instant is equal to



(A) $\frac{\pi r^2}{R}$ (B) $\frac{10r^3}{R}$

(C) $\frac{\pi^2 r^4 R}{5}$ (D) $\frac{\pi^2 r^4}{4R}$

7. In a YDSE apparatus, two films of thickness t_A and t_B, having refractive indices μ_A and μ_B are placed in front of slits A and B respectively. If $t_A\mu_A = t_B\mu_B$ the central maximum will

(A) not shift (B) shift towards A if $t_A < t_B$

(C) shift towards B if $t_A < t_B$ (D) shift towards A if $t_A > t_B$

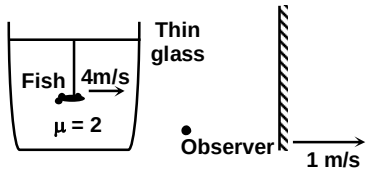
8. Velocity of image of fish in the mirror with respect to ground is

(A) zero

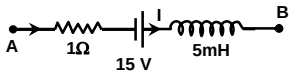
(B) 2 m/s towards mirror.

(C) 2 m/s away from mirror

(D) 4 m/s towards mirror.



9. The network shown in figure is a part of a complete circuit. If at a certain instant the current (I) is 5A, and is decreasing at a rate of 10^3 A/s. Then $V_B - V_A$ is equal to



(A) 15V (B) 25V

(C) 20V (D) 10 V

10. A planoconvex lens made of a material of refractive index 1.5 is silvered on the convex surface. Calculate the radius of curvature of the curved surface (Given : silvered lens which behaves as a mirror of focal length of -20 cm):

(A) 10 cm (B) 20 cm

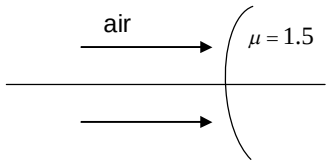
(C) 30 cm (D) 60 cm

Space for rough work

Multiple Correct Choice Type

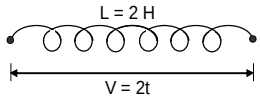
This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. Parallel rays of light are falling on convex spherical surface of radius of curvature R = 20 cm. Refractive index of the medium is $\mu = 1.5$. After refraction from the spherical surface parallel rays :



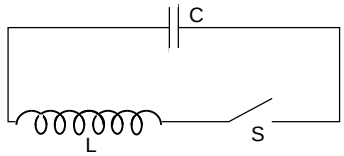
(A) actually meet at some point
(B) appears to meet after extending the refracted rays backwards
(C) meet (or appears to meet) at a distance of 30 cm from the spherical surface
(D) meet (or appears to meet) at a distance of 60 cm from the spherical surface

12. A variable voltage $V = 2t$ is applied across an inductor of inductance $L = 2H$ as shown in figure. Then,



(A) current versus time graph is a parabola
(B) energy stored in magnetic field at t = 2s is 4J
(C) potential energy at time t = 1 s in magnetic field is increasing at a rate of 1 J/s
(D) energy stored in magnetic field is zero all the time.

13. In the given circuit, initially the charge on capacitor is Q_o. At t = 0, the switch S is closed. Which of the following statement is/are correct?



(A) Maximum current through the inductor is $\frac{Q_o}{\sqrt{LC}}$

(B) Charge on the capacitor is zero at $t = \frac{\pi}{2}\sqrt{LC}$

(C) Current through the inductor is aero at $t = \frac{\pi}{2}\sqrt{LC}$

(D) Energy stored in magnetic field is zero all the time.

14. The potential difference applied to an X-ray tube is increased. Then which of the following is incorrect.
(A) The intensity increases.
(B) The minimum wavelength increases.
(C) The intensity decreases.
(D) The minimum wavelength remains constant.

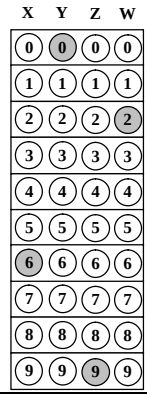
15. Which of the following is correct?
(A) Young's double slit experiment uses in coherent source
(B) Incoherent sources may having variable phase difference.
(C) Young's double slit experiment uses in incoherent source
(D) Incoherent sources must having same phase difference.

Space for rough work

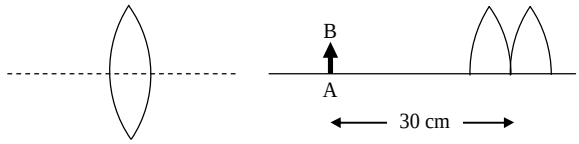
SECTION – C

Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. If a thin convex lens of focal length 20 cm is cut into two pieces from a plane passing through the pole, then both the pieces (touching each other) kept as shown in figure. If a object AB of length 2 cm is kept at distance of 30 cm. Find the height (in cm) of the image.

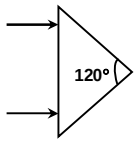


2. A solenoid of inductance 100 mH and resistance 20 Ω is connected to a cell of emf 10 V at t = 0. If the energy stored in the inductor when the time t = 5 ln2 milliseconds is $\frac{1}{64x}$ J, then find the value of "x"?

3. A wire of length ℓ = 6 ± 0.06 cm and radius r = 0.5 ± 0.005 cm and mass m = 0.3 ± 0.003 gm. Then find the maximum percentage error in density.

4. If the attractive potential between electron and proton were given by $V(r) = V_0 \ln\left(\frac{r}{r_0}\right)$ (where V_0, r_0 are constants), the radius r_n of the nth Bohr orbit in Hydrogen atom would be proportional to n^K. Then find the value of K.

5. An isosceles prism of angle 120°C has a refractive index 1.44. Two parallel monochromatic rays enter the prism parallel to each other in air as shown. The rays emerge from the opposite faces make an angle **K** [$\sin^{-1}(0.72) - 30°$] with each other. Then find the value of K.



Space for rough work

Chemistry

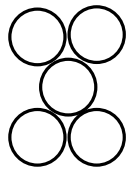
PART – II

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The packing efficiency of the two dimensional square unit cell shown below is



(A) 39.27 %
(B) 68.02%
(C) 74.05 %
(D) 78.54 %

2. The freezing point of aqueous solution contains 5% by mass urea, 1% by mass KCl and 10% by mass of glucose is (K_f (H_2O) = 1.86 K molality)
(A) 290.2 K
(B) 295.5 K
(C) 269.9 K
(D) 250 K

3. Zn gives H_2 gas with H_2SO_4 and conc. HCl but not with conc. HNO_3 because
(A) NO_3^- ion is reduced in preference to hydronium ion
(B) conc. HNO_3 is a weaker acid than conc H_2SO_4 and conc. HCl
(C) conc. HNO_3 acts as a reducing agent
(D) Zn is more reactive than H_2

4. $\left[Fe(H_2O)_6\right]^{2+}$ and $\left[Fe(CN)_6\right]^{4-}$ differ in
(A) magnetic moment and colour
(B) geometry and magnetic moment
(C) oxidation state of central metal ion and number of unpaired electron
(D) geometry and hybridization

5. $HClO_4 + P_2O_5 \longrightarrow A + B$
(A) and (B) are
(A) $HClO_3$, H_3PO_4
(B) Cl_2O_6, HPO_3
(C) Cl_2O_7, HPO_3
(D) ClO_2, H_3PO_4

Space for rough work

6. Which of the following minerals does not contain aluminium?
(A) Cryolite (B) Mica
(C) Feldspar (D) Fluorspar

7. Copper metal on treatment with dil HNO_3 produces a gas X. X when passed through acidic solution of stannous chloride a nitrogen containing compound Y is obtained. Y on reaction with nitrous acid produces a gas Z. Gas Z is
(A) NO (B) N_2
(C) NO_2 (D) N_2O

8. Alum helps in purifying water by
(A) Coagulating the mud particles
(B) Making mud water soluble.
(C) Forming Si complex with clay particles.
(D) Sulphate part which coming with dirt and remove it.

9. Which of the following order is correct for bond angles in given compound?

OF_2	OCl_2	OH_2
(I)	(II)	(III)

(A) I > II > III (B) III > I >II
(C) II > III > I (D) II > I > III

10. Which of the following complex is sp^3 hybridized and paramagnetic?
(A) $Ni(CO)_4$ (B) $[Ni(CN)_4)]^{2-}$
(C) $[NiCl_4]^{2-}$ (D) $[Cu(NH_3)_4]^{2+}$

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. Which of the following compounds are coloured due to charge transfer spectrum?
(A) K_2CrO_7 (B) $KMnO_4$
(C) $[Co(NH_3)_6]Cl_3$ (D) $[Cu(NH_3)_4]SO_4$

12. Nitrogen dioxide can be obtained by heating
(A) KNO_3 (B) $Pb(NO_3)_2$
(C) $LiNO_3$ (D) $AgNO_3$

13. Which of the following statement is correct?
(A) Fe^{2+} gives brown colour with ammonium thiocyanate
(B) Fe^{2+} gives blue precipitate with potassium ferricyanide
(C) Fe^{3+} gives brown colour with potassium ferricyanide
(D) Fe^{3+} gives red colour with potassium ferrocyanide

14. Which of the following are ambidentate ligands?
(A) CN^- (B) SCN^-
(C) NO_2^- (D) CH_3COO^-

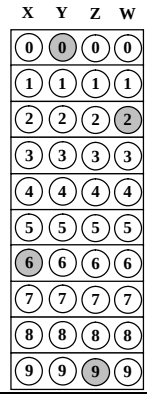
15. In which of the following pairs of solutions will be the values of the Van't Hoff factor be the same
(A) 0.05 M $K_4[Fe(CN)_6]$ and 0.10 M $FeSO_4$
(B) 0.10 M $K_4[Fe(CN)_6]$ and 0.05 M $FeSO_4$. $(NH_4)_2.SO_4.6H_2O$
(C) 0.20 M NaCl and 0.l M $BaCl_2$
(D) 0.05 M $FeSO_4(NH_4)_2SO_46H_2O$ and 0.02 M $KCl.MgCl_2.6H_2O$

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. The total number of P – O – P and P – O – H bonds present in pyro phosphoric acid.

2. How many lone pairs are present in H_3O^+?

3. How many bond angles are 90°C in XeF_4?

4. Number of salts among the following that will give metal on heating is $Zn(NO_3)_2$, $AgNO_3$, $Cu(NO_3)_2$, $Hg(NO_3)_2$.

5. What is the value of spin only magnetic moment value of $Fe(CO)_5$?

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Solution of the differential $y' = \frac{3yx^2}{x^3 + 2y^4}$ is

(A) $x^3y^{-1} = \frac{2}{3}y^3 + c$ (B) $x^2y^{-1} = \frac{2}{3}y^3 + c$

(C) $xy^{-1} = \frac{2}{3}y^3 + c$ (D) none of these

2. Let $f(x) = \begin{vmatrix} [x] & \{x\} & 1 \\ \{x\} & 1 & [x] \\ 1 & [x] & \{x\} \end{vmatrix}$, where [x] = greatest integer function and {x} = fractional part of x.

Then which of the following is incorrect

(A) $f(x) \geq 0$ for $x \in (-\infty, -1]$

(B) f(x) has no real root for $x \in (0,1)$

(C) for $x \in [0,1)$, range of f(x) is $(-2, -1]$

(D) for $x \in [1, 2)$, f(x) has exactly one local minima

3. The solution of $\frac{xdy}{x^2+y^2} = \left(\frac{y}{x^2+y^2} - 1\right)dx$ is

(A) $y = x\cot(c - x)$ (B) $\cos^{-1}\left(\frac{y}{x}\right) = -x + C$

(C) $y = x\tan(C - x)$ (D) $\frac{y^2}{x^2} = x\tan(C - x)$

Space for rough work

4. Value of $\begin{vmatrix} \cos 2\pi/63 & \cos 3\pi/90 & \cos 4\pi/77 \\ \cos(\pi/72) & \cos \pi/40 & \cos 3\pi/88 \\ 1 & \cos(\pi/90) & \cos(2\pi/99) \end{vmatrix}$ is equal to

(A) 0 (B) –1
(C) –2 (D) none of these

5. If $\Delta = \begin{vmatrix} \sin A & \sin B & \sin C \\ \cos A & \cos B & \cos C \\ \cos^3 A & \cos^3 B & \cos^3 C \end{vmatrix} = 0$ and A, B, C are angles of a triangle

Then
(A) ΔABC is always isosceles (B) ΔABC is always equilateral
(C) ΔABC is always scalable (D) none of these

6. Solution of $ydx - xdy + (1+x^2)dx + x^2 \sin y dy = 0$

(A) $\frac{y}{x} + \frac{1}{x} - x + \cos y + c = 0$ (B) $\frac{y}{x} + \frac{1}{x^2} - x + \cos y + c = 0$

(C) $\frac{2y}{x} - \frac{1}{x^2} - x + \cos y + c = 0$ (D) none of these

7. The simplification of $\begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix}^{15}$ is equal to

(A) $-2^{15}I$ (B) $2^{15}I$
(C) $(\sqrt{3})^{15}I$ (D) none of these

8. If A and B are two 3 × 3 symmetric matrix such that X = AB – BA then det(X) is equal to
(A) 0 (B) 1
(C) 2 (D) 3

9. The curve whose equation satisfies $x\frac{dy}{dx} - 4y = x^2\sqrt{y}$ passes through $(1,(\ell n2)^2)$, then the value of y(2) is
(A) $6\,\ell n2$ (B) $(2\,\ell n2)^2$
(C) $(6\,\ell n2)^2$ (D) none of these

Space for rough work

10. The probability of scoring a century in a cricket match by Sachin and Saurabh are $\frac{1}{4}$ and $\frac{1}{5}$ respectively and probability of getting out of a player by bowled, LBW, Catch, Run and stumping are $\frac{1}{7},\frac{2}{7},\frac{2}{7},\frac{1}{7}$ and $\frac{1}{7}$ respectively, if Sachin score a century while Saurabh does not, then the probability Sachin is catch out and Saurabh is bowled, is (given both players got out)

(A) $\frac{4}{49}$ (B) $\frac{1}{19}$

(C) $\frac{2}{49}$ (D) $\frac{2}{19}$

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. For fixed integer n, $\Delta = \begin{vmatrix} n! & (n+1)! & (n+2)! \\ (n+1)! & (n+2)! & (n+3)! \\ (n+2)! & (n+3)! & (n+4)! \end{vmatrix}$, then $\left[\frac{\Delta}{(n!)^3} - 4\right]$ is divisible by

(A) n (B) n^2+4n+5

(C) n^2+4n-5 (D) $2n^2$

12. $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & -3 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $6A^{-1} = aA^2 + bA + cI$, where a, b, c $\in$ integers and I is identity matrix. Then

(A) a + b + c = – 2 (B) $A^{-1} = \frac{1}{6}\begin{bmatrix} -3 & 1 & 6 \\ 0 & -2 & 0 \\ 3 & 1 & 0 \end{bmatrix}$

(C) $a+b+c=18$ (D) $A^{-1} = \frac{1}{12}\begin{bmatrix} 3 & 3 & 2 \\ 2 & 4 & -1 \\ 5 & 1 & -4 \end{bmatrix}$

Space for rough work

13. If E and F are two independent events such that $P(E \cap F) = \frac{k}{3}$ and $P(\bar{E} \cap \bar{F}) = \frac{k}{6}$ $(k \in I)$ and $(P(E) - P(F))(1 - P(F)) > 0$. Then

(A) $P(E) = \frac{2}{3}$ (B) $P(F) = \frac{1}{2}$

(C) $P(E) = \frac{1}{2}$ (D) $P(E) + P(F) = \frac{7}{6}$

14. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar unit vectors also $\vec{b}$, $\vec{c}$ are non-collinear and $2\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \vec{c}$, then

(A) angle between $\vec{a}$ and $\vec{c}$ is 60° (B) angle between $\vec{b}$ and $\vec{c}$ is 30°

(C) angle between $\vec{a}$ and $\vec{b}$ is 120° (D) $\vec{b}$ is perpendicular to $\vec{c}$

15. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar vectors such that $\bar{a}.\bar{a} = \bar{b}.\bar{b} = \bar{c}.\bar{c} = 3$ and $\bar{a}.\bar{b} = \bar{b}.\bar{c} = \bar{c}.\bar{a} = 1$, then

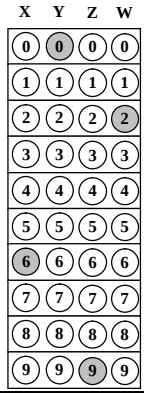
(A) $\left[\bar{a}+\bar{b} \quad \bar{b}+\bar{c} \quad \bar{c}+\bar{a}\right] = 4\sqrt{5}$ (B) volume of tetrahedron is $\frac{\sqrt{5}}{3}$

(C) $\left[\bar{a}\times\bar{b} \quad \bar{b}\times\bar{c} \quad \bar{c}\times\bar{a}\right] = 20$ (D) $\left[\bar{a} \ \bar{b} \ \bar{c}\right] = \frac{\sqrt{5}}{3}$

SECTION – C

Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. Three boxes are labelled as X, Y, Z and each box contains 5 balls numbered 1, 2, 3, 4, 5. The balls in each box are well mixed. One ball is chosen at random from each of the three boxes X, Y, Z respectively. If α, β, γ are the numbers on the balls chosen, then the probability that $\alpha = \beta + \gamma$ is equal to $\frac{K}{25}$ then K is equal to _____.

Space for rough work

2. There are 3 different pairs (i.e. 6 units say a, a, b, b, c, c) of shoes in a lot. Now three persons come and pick the shoes randomly (each gets 2 units). Let p be the probability that no one is able to wear shoes (i.e. no one gets a correct pair), then the value of $\frac{13p}{4-p}$, is _____.

3. Let $\vec{a}=\vec{b}+\vec{c}, \vec{b}\times\vec{d}=\vec{0}, \vec{c}.\vec{d}=0, |\vec{d}|\neq 0$ and $|\vec{c}|=2$ then the value of $\frac{\left|\vec{d}\times\left(\vec{a}\times\vec{d}\right)\right|}{\left|\vec{d}\right|^2|\vec{c}|}$ is equal to _____.

4. Let $\vec{a}=\cos\theta\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}+\cos\theta\hat{j}+\hat{k}, \vec{c}=\hat{i}+\hat{j}+\hat{k}$, $\left|\vec{a}.(\vec{b}\times\vec{c})\right|$ is maximum, then square of the shortest distance between $\vec{r}=\vec{a}+\lambda_1\vec{b}$ and $\vec{r}=\vec{b}+\lambda_2\vec{c}$ is _____.

5. The differential equation of $\left(y\sqrt{x^2+y^2}-x\right)dx+\left(x\sqrt{x^2+y^2}-y\right)dy=0$ represents a curve given by $f(x, y)=0$. The point (0, 2) satisfies the equation of the curve and the value of x when y = 1 is $\frac{-h}{4}$. Then h is equal to _____.

Space for rough work